PRADUMN SINGH

Deep Learning Researcher — Machine Learning Engineer — Generative AI & Agentic Systems

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Professional Summary

Electrical and Electronics Engineering sophomore at NIT Tiruchirappalli with a strong **growth mindset** and passion for **Computer Science and Machine Learning**, and a strong foundation in **object-oriented programming**, and **data structures and algorithms**. Experienced in **collaborative development**, **feature design and implementation**, and applying **problem-solving skills** across the **development lifecycle** while working on real-world AI-driven systems.

Education

National Institute of Technology, Tiruchirappalli — Aug 2024 – Present
B.Tech in Electrical and Electronics Engineering – ***CGPA: 9.57/10.0***

City Montessori School — 2017 – 2024
Class 12 (ISC): ***96%*** | *Class 10 (ICSE):* ***97%***

Standardized Tests: JEE MAINS - **99.579 percentile (Top 0.421% among 1.4M+ students)** | SAT - **1450/1600 (800/800 in Mathematics)**| IELTS - **8.0/9.0**

Experience

Research Intern — Nov 2025 – Present
Predictive Maintenance & Reinforcement Learning Research

- Engineered a hybrid **Transformer-PPO** architecture for **100 heterogeneous engines**, achieving a state-of-the-art **L1 Loss of 6.0** on the CMAPSS dataset.
- Utilized a **50-cycle** temporal window and **26 sensor channels** to capture degradation, reducing RUL prediction error by **30%** over LSTM baselines via **multi-head attention**.
- Implemented a **Proximal Policy Optimization** agent with an **8-dimensional state vector** and **3-point** discrete action space to optimize scheduling using **deep reinforcement learning**.

AI/ML Developer — Aug 2025 – Present
SPIDER R&D, NIT Tiruchirappalli

- Exhibit **teamwork** in an Agile environment to design, and train various AI models using version control and reproducible data pipelines.
- Benchmark performance across architectures to optimize deployment and conduct research on **INR for audio super-resolution**.

Workshops Coordinator — Oct 2025 – Present
Pragyan (NIT Trichy's International Tech Fest)

- Demonstrate **leadership** by pitching collaboration opportunities to industry representatives to secure corporate partnerships in technical workshops.
- Coordinate **multi-stakeholder engagements** and team efforts to manage logistics for the international technical festival.

Key Projects

LynxGPT – RAG-Based Educational Assistant
SPIDER R&D | Sep 2025 – Dec 2025

- Following the **Software Development Lifecycle (SDLC)** Designed and Developed end-to-end **Retrieval Augmented System system** with custom scraper aggregating **2,000+ documents** into Supabase vector DB **(dim 384)**
- Utilized LangGraph to construct multi-agent orchestrator routing queries across **3-agent architecture** achieving **80% response accuracy**
- Optimized retrieval with **top-6 similarity search** and **12-message sliding window** for conversational memory
- Deployed production stack: **React frontend, FastAPI backend, MongoDB for session management**

YOLOv1 in Pytorch
Computer vision Project
- Developed **YOLOv1** architecture trained on **2,800+ training images** to detect **10 classes** of construction safety gear.
- **Stopped class imbalance** by implementing a **weighted loss function** and a custom background penalty to stabilize the **0.60 confidence value** and minimize false positives.
- Optimized detection accuracy using a tuned **0.10 IoU threshold** for NMS and conducted **visual analysis on 10 randomly chosen images** to ensure bounding box alignment.

Autonomous Market Research Agent – TRANSFINITE Hackathon
Runner-Up (100+ teams) | NIT Tiruchirappalli
- Collaborative Development of an autonomous qualitative research agent using **LangChain, LangGraph** with **RAG 2.0** and **few-shot prompting** for businesses
- Engineered agentic state machine with **3-point discrete action space** achieving **5-6 layers of recursive probing**
- Maintained coherent, human-like dialogue flow for deep consumer interviews without manual intervention

WGAN-GP for image Generation
Personal Project
- Developed **WGAN-GP** architecture synthesizing high-fidelity **64x64 images** from **60,000+ training samples**, achieving **FID of 24**
- Implemented gradient penalty for **1-Lipschitz continuity** ensuring convergence stability, debugging and Code Optimization to improve efficiency.
- Deployed real-time **web application** using Streamlit for instantaneous generation from **100-dimensional latent space**

Multi-Agent Surgical Safety System – OMRL X PRAGYAN 2nd Place
2nd Place (20+ teams) | NIT Tiruchirappalli
- Performed comprehensive **Requirements Analysis** to align technical architecture with clinical safety needs and surgical workflows.
- Built a real-time **Multi-Agent Orchestrator** utilizing **Audio Diarization** for speaker identification, achieving **90% accuracy** in monitoring surgical safety protocols.
- Architected a **3-agent Gen-AI system** (Checklist Generator/Marker, Form Filler, and Automated Summarizer) to process and analyze **live Operating Room (OR) conversations**.
- Deployed a production-ready pipeline using **LangChain, LangGraph, and Flask** to provide real-time updates and automated documentation for surgical protocols.

Skills

AI/ML Specializations: Agentic AI, Reinforcement Learning, Generative Networks, Deep Learning, RAG Systems, YOLO

Programming Languages: Python, Java, C++, C, SQL, JavaScript, Cypher

ML Frameworks & Tools: PyTorch, LangChain, LangGraph, Gymnasium, Scikit-learn, HuggingFace

Development Tools: FastAPI, React.js, Version Control (Git), GitHub, MongoDB, Supabase, PostgreSQL, Neo4j

Libraries: NumPy, Pandas, OpenCV, Matplotlib, Pytesseract, Psycopg2

Other Software: Matlab, LTSpice, Unity, Visual Studio Code, PowerPoint, Canvas

Soft Skills: Communication, Leadership, Problem-solving, Team Collaboration

Certifications

- Convolutional Neural Networks - By DeepLearning.AI
- Data Analytics with Python - By IBM
- Data Science Methodology - By IBM

Languages

- **English** : Native level fluency, band 8.0 IELTS
- **Hindi** : Native level fluency